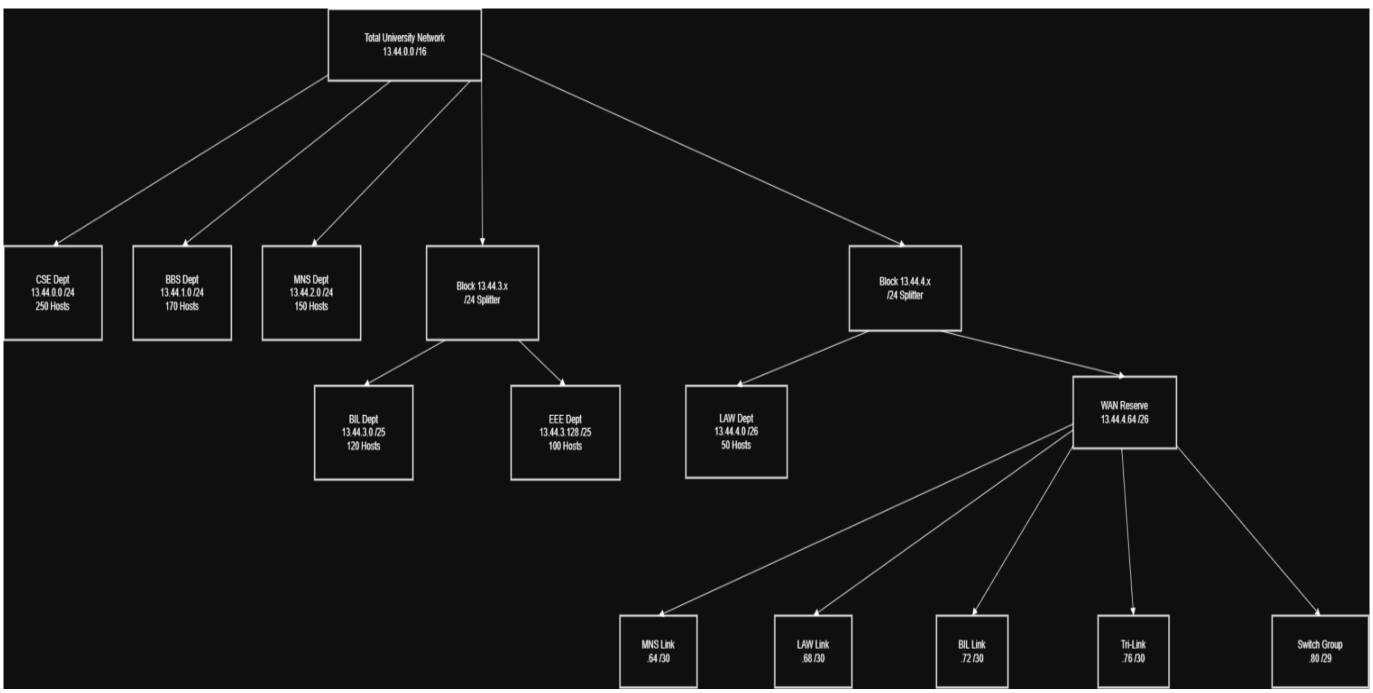
IP ADDRESS TABLE

Device Name	Interface	IP Address	Subnet Mask	Default Gateway
CSE Router	Gig0/0	13.44.0.1	255.255.255.0	N/A
	Gig0/1	13.44.4.81	255.255.255.248	N/A
	Se0/0/0	13.44.4.65	255.255.255.252	N/A
	Se0/0/1	13.44.4.69	255.255.255.252	N/A
	Se0/1/0	13.44.4.73	255.255.255.252	N/A
BBS Router	Gig0/0	13.44.1.1	255.255.255.0	N/A
	Gig0/1	13.44.4.83	255.255.255.248	N/A
MNS Router	Gig0/0	13.44.2.1	255.255.255.0	N/A
	s0/0/0	13.44.4.66	255.255.255.252	13.44.4.65
BIL Router	Gig0/0	13.44.3.1	255.255.255.128	N/A

	Se0/0/0	13.44.4.74	255.255.255.252	N/A
	Se0/0/1	13.44.4.78	255.255.255.252	N/A
EEE Router	Gig0/0	13.44.3.129	255.255.255.128	N/A
	Gig0/1	13.44.4.82	255.255.255.248	N/A
LAW Router	Gig0/0	13.44.4.1	255.255.255.192	N/A
	Se0/0/0	13.44.4.70	255.255.255.252	N/A
	Se0/0/1	13.44.4.77	255.255.255.252	N/A
PC CSE	NIC	13.44.0.10	255.255.255.0	13.44.0.1
PC BBS	NIC	13.44.1.10	255.255.255.0	13.44.1.1
PC MNS	NIC	DHCP	DHCP	DHCP
PC EEE	NIC	DHCP	DHCP	DHCP
PC LAW	NIC	DHCP	DHCP	DHCP

PC BIL	NIC	DHCP	DHCP	DHCP

VLSM TREE



Configuration commands

#R_CSE

R_CSE>en
R_CSE#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R_CSE(config)#hostname R_CSE
R_CSE(config)#no ip domain-lookup
R_CSE(config)#int gig0/0
R_CSE(config-if)#ip address 13.44.0.1 255.255.255.0
R_CSE(config-if)#no shut

R_CSE#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R_CSE(config)#int gig0/1
R_CSE(config-if)#ip address 13.44.4.81 255.255.255.248
R_CSE(config-if)#no shut

R_CSE#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R_CSE(config)#int s0/0/0
R_CSE(config-if)#ip address 13.44.4.65 255.255.255.252
R_CSE(config-if)#clock rate 64000
R_CSE(config-if)#no shut
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down

R_CSE(config-if)#int s0/0/1
R_CSE(config-if)#ip address 13.44.4.69 255.255.255.252
R_CSE(config-if)#clock rate 64000
R_CSE(config-if)#no shut
%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down

R_CSE(config-if)#int s0/1/0
R_CSE(config-if)#ip address 13.44.4.73 255.255.255.252
R_CSE(config-if)#clock rate 64000
R_CSE(config-if)#no shut

```
R_CSE>en
R_CSE#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R_CSE(config)#ip dhcp excluded-address 13.44.0.1 13.44.0.20
R_CSE(config)#ip dhcp excluded-address 13.44.1.1 13.44.1.20
R_CSE(config)#ip dhcp excluded-address 13.44.2.1 13.44.2.20
R_CSE(config)#ip dhcp excluded-address 13.44.3.129 13.44.3.140
R_CSE(config)#ip dhcp pool CSE_POOL
R_CSE(dhcp-config)#network 13.44.0.0 255.255.255.0
R_CSE(dhcp-config)#default-router 13.44.0.1
R_CSE(dhcp-config)#dns-server 13.44.0.2
R_CSE(dhcp-config)#exit
R_CSE(config)#ip dhcp pool BBS_POOL
R_CSE(dhcp-config)#network 13.44.1.0 255.255.255.0
R_CSE(dhcp-config)#default-router 13.44.1.1
R_CSE(dhcp-config)#dns-server 13.44.0.2
R_CSE(dhcp-config)#exit
R_CSE(config)#ip dhcp pool MNS_POOL
R_CSE(dhcp-config)#network 13.44.2.0 255.255.255.0
R_CSE(dhcp-config)#default-router 13.44.2.1
R_CSE(dhcp-config)#dns-server 13.44.0.2
R_CSE(dhcp-config)#exit
R_CSE(config)#ip dhcp pool EEE_POOL
R_CSE(dhcp-config)#network 13.44.3.128 255.255.255.128
R_CSE(dhcp-config)#default-router 13.44.3.129
R_CSE(dhcp-config)#dns-server 13.44.0.2
R_CSE(dhcp-config)#exit

R_CSE>en
R_CSE#conf t
R_CSE(config)#router rip
R_CSE(config-router)#version 2
R_CSE(config-router)#network 13.44.0.0
R_CSE(config-router)#network 13.44.1.0
R_CSE(config-router)# network 13.44.3.128
R_CSE(config-router)#network 13.44.4.80
R_CSE(config-router)#no auto-summary
R_CSE(config-router)#exit
R_CSE(config)#end
```

R_CSE#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R_CSE(config)#router rip
R_CSE(config-router)#version 2
R_CSE(config-router)#network 13.44.0.0
R_CSE(config-router)#network 13.44.4.80
R_CSE(config-router)#no auto-summary
R_CSE(config-router)#exit
R_CSE(config)#ip route 13.44.2.0 255.255.255.0 13.44.4.66
R_CSE(config)#ip route 13.44.3.0 255.255.255.128 13.44.4.74
R_CSE(config)#ip route 13.44.4.0 255.255.255.192 13.44.4.70
R_CSE(config)#end

#R_MNS

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R_MNS
R_MNS(config)#no ip domain-lookup

R_MNS(config)#int g0/0
R_MNS(config-if)#ip address 13.44.2.1 255.255.255.0
R_MNS(config-if)#no shut

R_MNS(config-if)#int s0/0/0
R_MNS(config-if)#ip address 13.44.4.66 255.255.255.252
R_MNS(config-if)#no shut

R_MNS>en
R_MNS#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R_MNS(config)#interface GigabitEthernet0/0
R_MNS(config-if)#ip address 13.44.2.1 255.255.255.0
R_MNS(config-if)#ip helper-address 13.44.4.81

R_MNS(config-if)#no shutdown
R_MNS(config-if)#exit
R_MNS# conf t
R_MNS(config)#ip route 0.0.0.0 0.0.0.0 13.44.4.65
R_MNS(config)#end

#R_BBS

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R_BBS
R_BBS(config)#no ip domain-lookup
R_BBS(config)#int g0/0
R_BBS(config-if)#ip address 13.44.1.1 255.255.255.0
R_BBS(config-if)#no shut
R_BBS(config-if)#exit
R_BBS(config)#int g0/1
R_BBS(config-if)#ip address 13.44.4.83 255.255.255.248
R_BBS(config-if)#no shut

R_BBS>en
R_BBS#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R_BBS(config)#interface GigabitEthernet0/0
R_BBS(config-if)#ip address 13.44.1.1 255.255.255.0
R_BBS(config-if)#ip helper-address 13.44.4.81
R_BBS(config-if)#no shutdown
R_BBS(config-if)#exit
R_BBS(config)#

R_BBS>en
R_BBS#conf t
Enter configuration commands, one per line. End with CNTL/Z.

R_BBS(config)#router rip
R_BBS(config-router)#version 2
R_BBS(config-router)#network 13.44.1.0
R_BBS(config-router)#network 13.44.4.80
R_BBS(config-router)#no auto-summary
R_BBS(config-router)#exit
R_BBS(config)#end

#R_EEE

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R_EEE
R_EEE(config)#no ip domain-lookup
R_EEE(config)#int g0/0
R_EEE(config-if)#ip address 13.44.3.129 255.255.255.128
R_EEE(config-if)#no shut
R_EEE(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R_EEE(config-if)#exit
R_EEE(config)#int g0/1
R_EEE(config-if)#ip address 13.44.4.82 255.255.255.248
R_EEE(config-if)#no shut
R_EEE(config-if)#end

R_EEE>en
R_EEE#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R_EEE(config)#interface GigabitEthernet0/0
R_EEE(config-if)#ip address 13.44.3.129 255.255.255.128
R_EEE(config-if)#ip helper-address 13.44.4.81
R_EEE(config-if)#no shutdown
R_EEE(config-if)#exit

R_EEE(config)#

R_EEE(config)#router rip
R_EEE(config-router)#version 2
R_EEE(config-router)#network 13.44.3.128
R_EEE(config-router)#network 13.44.4.80
R_EEE(config-router)#no auto-summary
R_EEE(config-router)#exit

#R_LAW

R_EEE(config-if)#end
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R_LAW
R_LAW(config)#no ip domain-lookup
R_LAW(config)#
R_LAW(config)#int g0/0
R_LAW(config-if)#ip address 13.44.4.1 255.255.255.192
R_LAW(config-if)#no shut
R_LAW(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
R_LAW(config-if)#exit
R_LAW(config)#int s0/0/0
R_LAW(config-if)#ip address 13.44.4.70 255.255.255.252
R_LAW(config-if)#no shut
R_LAW(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
R_LAW(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R_LAW(config-if)#exit
R_LAW(config)#int s0/0/1
R_LAW(config-if)#ip address 13.44.4.77 255.255.255.252
R_LAW(config-if)#no shut

R_LAW>en
R_LAW#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R_LAW(config)#ip dhcp excluded-address 13.44.4.1 13.44.4.20
R_LAW(config)#ip dhcp pool LAW_POOL
R_LAW(dhcp-config)#network 13.44.4.0 255.255.255.192
R_LAW(dhcp-config)#default-router 13.44.4.1
R_LAW(dhcp-config)#dns-server 13.44.0.2
R_LAW(dhcp-config)#exit
R_LAW(config)#

R_LAW(config)#ip route 0.0.0.0 0.0.0.0 Serial0/0/0
R_LAW(config)#ip route 0.0.0.0 0.0.0.0 Serial0/0/1 250
R_LAW(config)#ip route 13.44.3.0 255.255.255.128 Serial0/0/1
R_LAW(config)#end

#R_BIL

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R_BIL
R_BIL(config)#no ip domain-lookup
R_BIL(config)#int g0/0
R_BIL(config-if)#ip address 13.44.3.1 255.255.255.128
R_BIL(config-if)#no shut
R_BIL(config-if)#exit
R_BIL(config)#int s0/0/0
R_BIL(config-if)#ip address 13.44.4.74 255.255.255.252
R_BIL(config-if)#no shut
R_BIL#conf t
Enter configuration commands, one per line. End with CNTL/Z.

```
R_BIL(config)#int s0/0/1
R_BIL(config-if)#ip address 13.44.4.78 255.255.255.252
R_BIL(config-if)#clock rate 64000
R_BIL(config-if)#no shut


R_BIL>en
R_BIL#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R_BIL(config)#ip dhcp excluded-address 13.44.3.1 13.44.3.20
R_BIL(config)#ip dhcp pool BIL_POOL
R_BIL(dhcp-config)#network 13.44.3.0 255.255.255.128
R_BIL(dhcp-config)#default-router 13.44.3.1
R_BIL(dhcp-config)#dns-server 13.44.0.2
R_BIL(dhcp-config)#exit
R_BIL(config)#


R_BIL(config)#ip route 0.0.0.0 0.0.0.0 Serial0/0/0
R_BIL(config)#ip route 0.0.0.0 0.0.0.0 Serial0/0/1 250
R_BIL(config)#ip route 13.44.4.0 255.255.255.192 Serial0/0/1
R_BIL(config)#end
```

Assumptions:

1. Two PCs per department accurately represent all end devices in that department for testing and demonstration purposes.
2. RIP version 2 is used, since it supports VLSM and is appropriate for the given network design.
3. Static routing entries are correctly prioritized where required (LAW and BIL departments).
4. LAW and BIL have dedicated DHCP servers and do not rely on relay agents for IP assignment.
5. DNS resolution is centralized and all departments are correctly configured to use the CSE DNS server.
6. Printers are local to each department and do not require inter-department print services.
7. No additional security mechanisms (firewalls, ACLs, NAT) are required unless explicitly stated.
8. I assumed that all devices start in an operational state, with no boot-time failures or misconfigurations.
9. I assumed that testing via ping and basic service access is sufficient to verify full network functionality.
10. I assumed that the network design focuses on logical correctness, not real-world scalability or redundancy beyond what is specified.